

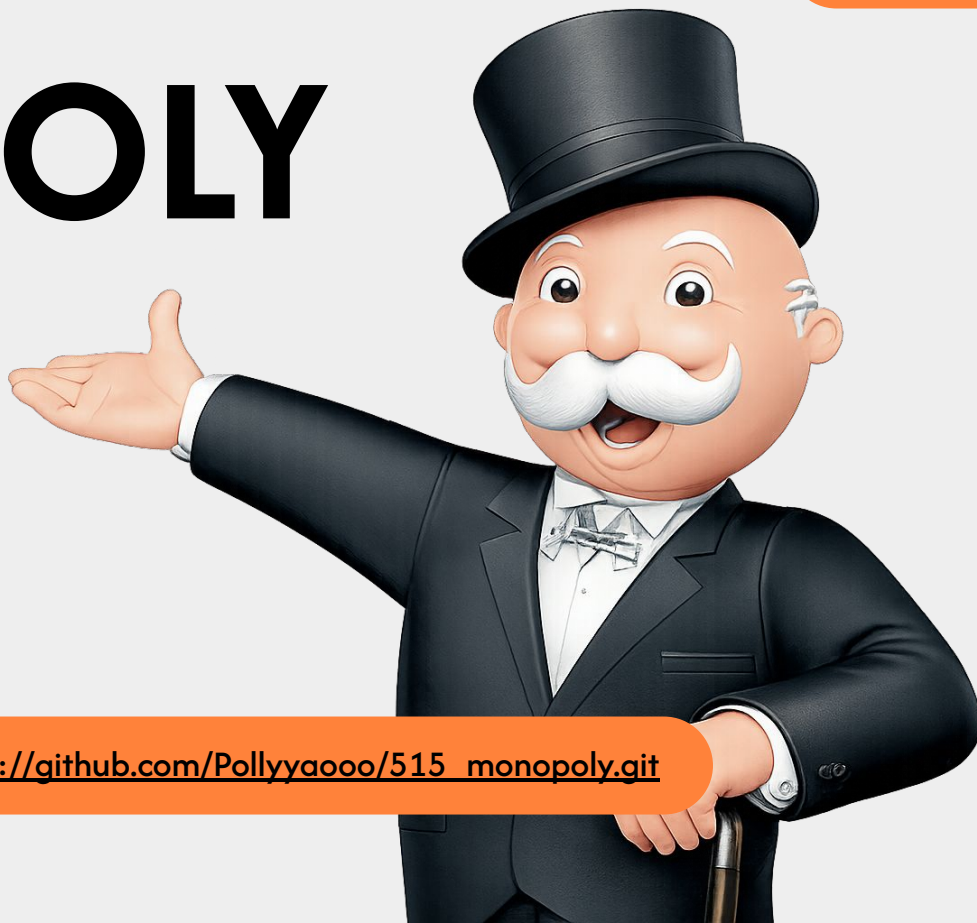
Morden MONOPOLY

Team 11

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https://github.com/Pollyyao000/515_monopoly.git





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How might we reduce the administrative burden of tracking game state and enforcing rules in Monopoly, so that all players can focus on strategy and fun?

Why we want...



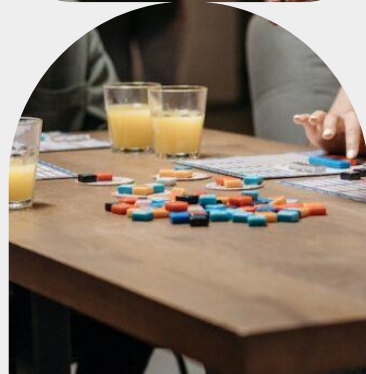
The problem

Monopoly requires players to manually track positions, property ownership, houses, rents, and card effects — creating cognitive overload, rule disputes, and slow gameplay.



Intended users

Anyone who plays Monopoly — from first-time players overwhelmed by rules, to experienced players who want faster, dispute-free games.



Solution

A smart Monopoly system that automatically tracks board state and enforces rules, so players never have to.

Board Intelligence

The physical board is enhanced with sensors to automatically detect and track:

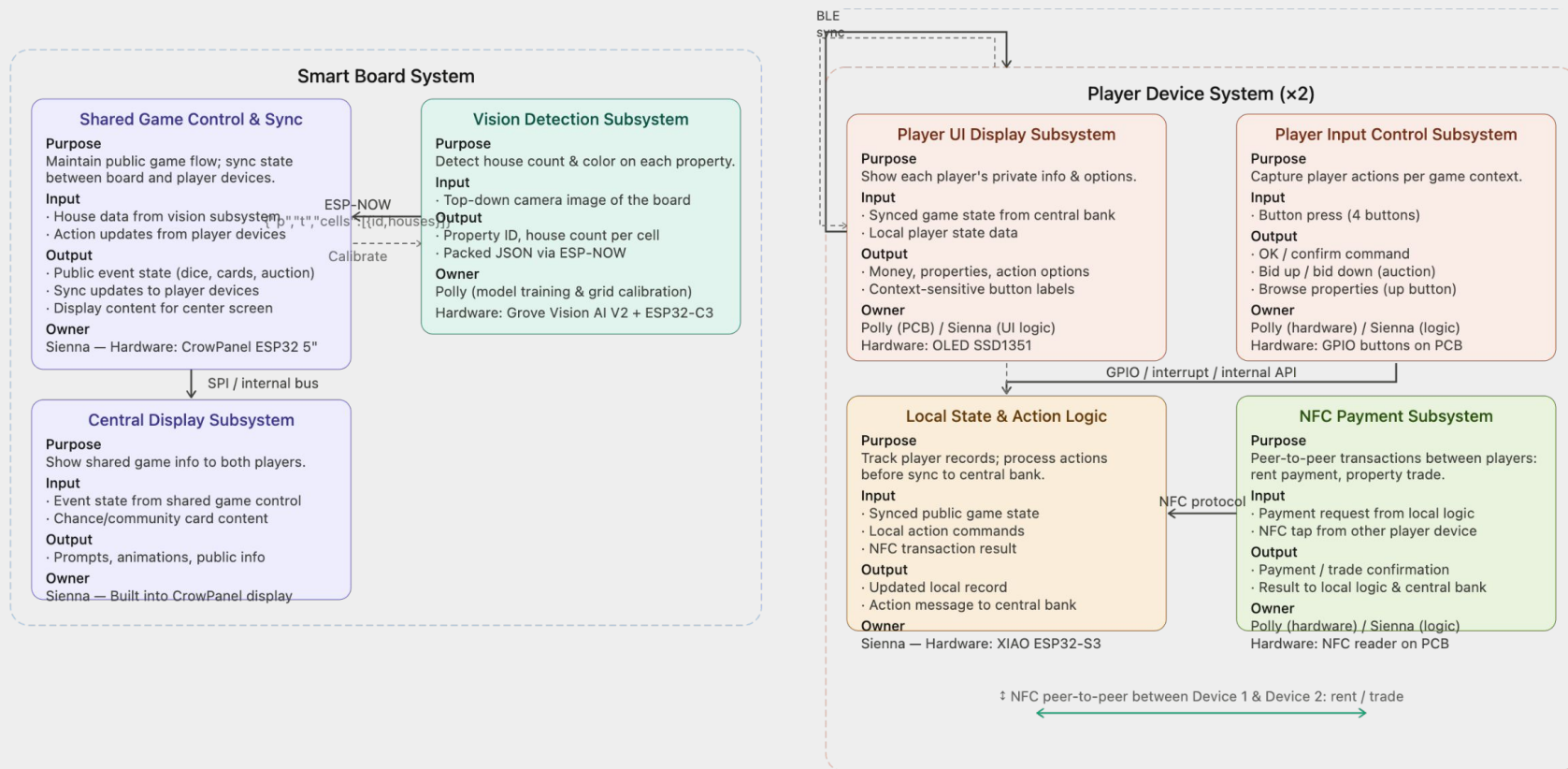
- Player token positions (who is where)
- Property ownership status
- Number and color of houses built on each property

Rules Intelligence

Game information is processed and displayed in real time across two interfaces:

- A Central Bank screen (public display) — shows card draws, rent calculations, property transactions...
- Individual player devices (private display) — shows personal balance, rent owed/collected, player-to-player transfers..

Architecture 2



Optimization 1 - Enclosure & Camera Mount Stability



	Before	After
Enclosure load capacity	Collapsed under its own use	Supports >1kg (laptop)
Camera mount stability	Visible wobble during use	Zero movement during testing



Optimization 2 - Central Display UI



Before

After

Information clarity

Basic, sparse layout

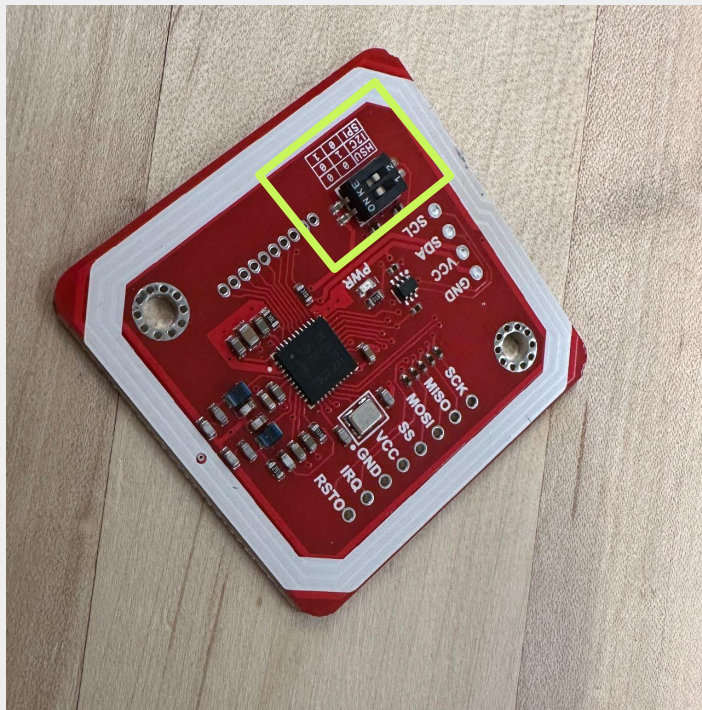
Full game state visible

Player guidance

Minimal prompts

Context-aware instructions

Optimization 3



	Before	After
Read distance	0mm (direct contact only)	~5mm (through 4mm of casing)
Protocol	Previous library	I2C via ONKE
Usability	Required removing casing	Works through full enclosure

Readiness - NFC vs auction



Condition 1: NFC-based rent payment

When a player lands on an opponent's property, the system triggers a rent payment flow. The paying player taps their device to the receiving player's NFC reader to complete the transaction, with the updated balance automatically synced to the central bank.

Condition 2: Button-based auction *When a property is not purchased by the landing player, an auction is triggered on the central display. Both players use the left/right buttons on their devices to raise or lower their bids, and confirm with the center button. The highest confirmed bid wins the property.*

Readiness - house vs hotel



Condition 1: Incorrect green house placement

A property is deliberately set up with an incorrect number of green houses on the board. The camera subsystem detects the mismatch between the physical board state and the expected game state, triggering a visual alert on the central display to prompt the players to correct the placement.



Condition 2: Incorrect red hotel placement

The same detection flow is tested with red hotels, which differ in both color and shape from green houses. The camera subsystem successfully identifies the incorrect number of hotels and triggers a corresponding alert on the central display, demonstrating that the vision model generalizes across both building types.

Budget update

A	B	C	D	E	F	G	H	I	J
BUDGET SUMMARY									
Team Num	11	CFO Name:	Polly Yao	Total Budget (\$):	\$350.00	Total Spent (\$):	\$165.55	Remaining (\$):	\$184.45
#	Item Name	Quantity	Unit Price (\$)	Total Price (\$)	Supplier	Link	Notes	Instructor Approval	Actual (\$)
1	A1324LUA-T	5	\$1.30	\$6.51	Digikey	https://www.digikey.com/en/products/detail/allegro-microsystems/A1324LUA-T/2728144		approve	\$15.24
2	Amphenol ICC (FCI) 609-2176-ND CONN FFC HEADER 2POS 2.54MM	15	\$0.74	\$11.04	Digikey	https://www.digikey.com/en/products/detail/amphenol-icc-fci-/66226-002LF/1002491?qcsrc=a		approve	\$20.04
3	waveshare 1.5inch RGB OLED Display Module, 128x128 Pixels Displaying 65K Colors, Support Raspberry Pi Arduino STM32, SPI Interface	2	\$21.93	\$43.86	Amazon	https://www.amazon.com/dp/B07DBXMFNS?ref=fed_asin_title		approve	\$65.58
4	ALMOCN 3PCS PN532 NFC NXP RFID Module V3 Kit Near Field Communication Reader Module Kit I2C SPI HSU with S50 White Card Key Card for Arduino Raspberry Pi DIY Smart Phone Android Phone	1	\$15.59	\$15.59	Amazon	https://www.amazon.com/dp/B09VC5BQFC?ref=fed_asin_title		approve	
5	200Pcs Tactile Push Button Switch Micro Momentary Tact Assortment Kit,10 Values 6x6mm 4 Pin	1	\$9.59	\$9.59	Amazon	https://www.amazon.com/dp/B09DKJ3C4K?psc=1&smid=A25GTVTP3XX1HF&ref=chk_tvp_i		approve	\$10.58
6	WYMCONGCONG 72 Pcs Momentary Tactile Push Button Micro Switch 6x6x9mm 6 Color Cap	1	\$10.59	\$10.59	Amazon	https://www.amazon.com/dp/B07F59CLHB?ref=fed_asin_title		approve	\$11.68
7	JST SH1.0 Connectors Horizontal SMD Socket and Crimp Terminals Kit Compatible with SH 1.0mm Pitch 2/3/4/5Pin Housing Adapter	1	\$9.49	\$9.49	Amazon	https://www.amazon.com/dp/B0FH9SZD66?ref=fed_asin_title		approve	\$10.47
8	Natural Hard Maple Veneer Wood Sheets - Ideal for DIY Speaker Renovation and Furniture Restoration - Unfinished Wooden Sticker - Handmade Style - 8 x 110 Inch	1	\$21.99	\$21.99	Amazon	https://www.amazon.com/dp/B0CWDJXBRR?ref=fed_asin_title		approve	\$24.25
9	20 Pcs NFC Tags Sticker NTAG215 Chip Anti-Metal NFC Stickers Adhesive Labels Transparent RFID Tags Rewritable 504 Bytes Memory Suitable for All NFC-Enabled	1	\$6.99	\$6.99	Amazon	https://www.amazon.com/dp/B0F2HZKCCW?ref=fed_asin_title		approve	\$7.71

Thank you!